



Rollins Tennis Archive

Madhav Khanal: Lead Programmer

Richard Stoiberer: Project Manager

Stella Fruijtier: Data Scientist/Data Visualization

Project Recap

Build a full-stack application that organizes **Rollins Tennis** player bios, match results, and statistics.

- Store players, matches, and seasons using OOP structure
- Allow filtering/searching of players and matches
- Show useful team and individual statistics

The goal is to create a streamlined platform that organizes historical player and match data in one accessible location, making it easier to review past performances than on the existing Rollins Tennis website.



Menu



WOMEN'S TENNIS

More +

2024-25 Women's Tennis Cumulative Statistics

View PDF

Select a Season...

SEASON STATS

MATCH-BY-MATCH

OVERALL

CONFERENCE

Singles

Abdulloeva, Amina

Eckstein, Sydney

Falster, Milla

Fruitier, Stella

Liu, Nancy

Mitrofanova, Nina

Sallick, Annie

Talbi, Amina

Vlasova, Polina

Total

Percentage

Doubles Teams

Falster, Milla / Fruitier, Stella

Falster, Milla / Liu, Nancy

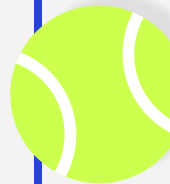
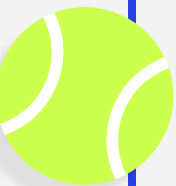
Falster, Milla / Mitrofanova, Nina

Falster, Milla / Vlasova, Polina

Liu, Nancy / Fruitier, Stella

Liu, Nancy / Vlasova, Polina

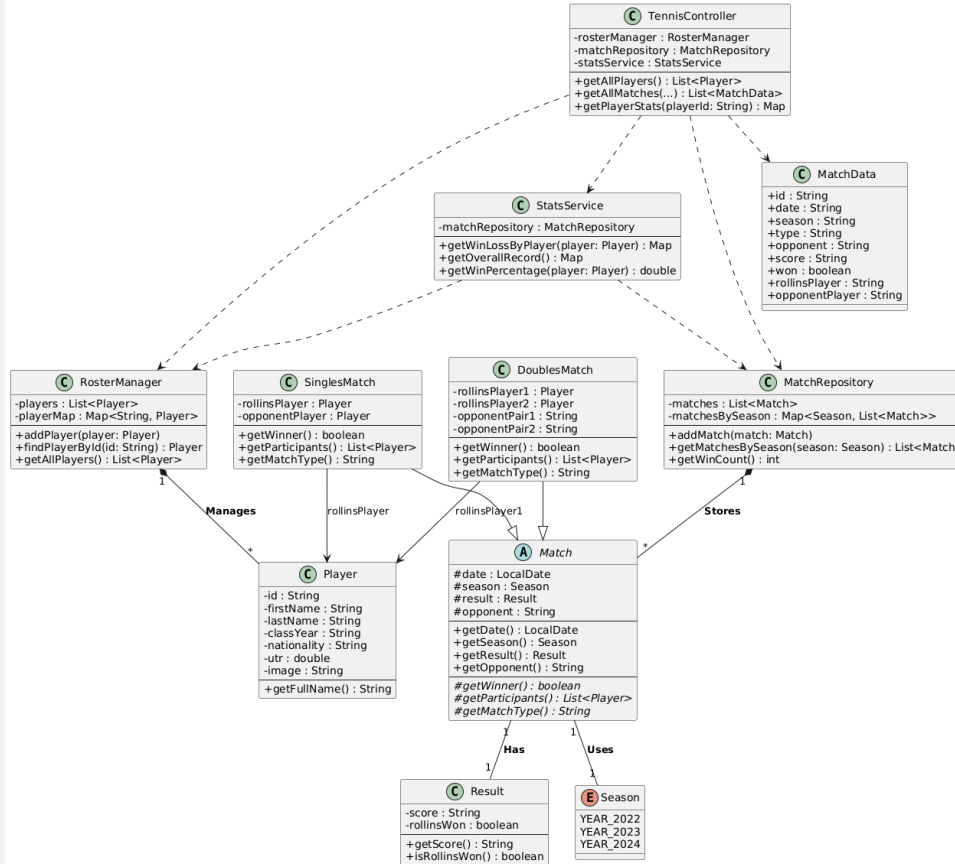
Demo



OOP Principles

- **Classes and Objects:** *Player, Match, SinglesMatch, DoublesMatch, Result, Season*
- **Encapsulation:** All fields are private and accessed through getter and setter methods
- **Inheritance:** Implemented through the *Match* class hierarchy
- **Abstraction:** *Match* is an abstract class with shared structure and behaviour
- **Polymorphism:** *SinglesMatch* and *DoublesMatch* both inherit from *Match*, allowing them to be stored and processed in the same collections

Rollins Tennis Archive - UML Class Diagram



UML Diagram

Challenges in the process

Challenges:

- Designing OOP structure working both for singles and doubles
- Organizing match data for efficient filtering and statistics
- Ensuring consistent data transfer between backend and frontend
- Creating meaningful statistics while keeping logic maintainable

Thank you 😊